

UVM reporting

Simulation logs get noisy fast, UVM reporting gives you two independent knobs

Severity, verbosity, and ID filter

- Use uvm info with an ID string, message text
- Uvm warning, uvm error, and uvm fatal set severity
- High verbosity info is filtered out at medium
- Set report id action and plusargs like UVM verbosity let you tune per run
- ID filters drop or demote matching IDs before print

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

UVM reporting ladder

Sketch how messages climb the ladder: **severity**, **verbosity**, and **ID filter**. Starter:
threshold UVM_MEDIUM, emit uvm_info("DRV", ..., UVM_LOW) — shown.

Starter example: threshold UVM_MEDIUM, emit uvm_info("DRV", "byte accepted", UVM_LOW) — shown.

Load starter example

Challenges 0 / 22 cleared

Quiz: severity: UVM report severity ladder (light → heavy) is...

☐ INFO → WARNING → ERROR → FATAL

☐ DEBUG → FULL → NONE only

☐ PASS → FAIL → SKIP

☐ build → connect → run

Show hint

Check

Next

Idle

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Core ideas

severity

INFO → WARNING → ERROR → FATAL gravity.

verbosity

How chatty INFO/WARNING may be.

threshold

Print if message verb ≤ configured level.

ID filter

Drop or demote by report ID string.

Lab

SCENARIO	THRESHOLD	SEVERITY	MSG VERB	ID
INFO DRV shown ▾	UVM_MEDIUM ▾	INFO ▾	UVM_LOW ▾	DRV

UVM reporting lab starter

Real UVM literacy

```
❯ module19-uvm-reporting/examples/reporting-sketch

# Track A -- real Unix
# real shell session (Track A -- UVM reporting)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_uvm2017/module19-uvm-reporting

$ ls examples/reporting-sketch/
reporting.txt

$ cat examples/reporting-sketch/reporting.txt
UVM reporting sketch (literacy)

Two independent knobs:
severity  = gravity of the message
verbosity = how chatty INFO/WARNING may be

Severity ladder (typical):
INFO      - routine status (non-fatal)
WARNING   - suspicious but continues
ERROR     - check failure; counts toward test fail
FATAL     - stops simulation

Verbosity ladder (typical):
UVM_NONE < UVM_LOW < UVM_MEDIUM < UVM_HIGH < UVM_FULL < UVM_DEBUG

Macros:
'uvm_info("DRV", "byte accepted", UVM_LOW)
'uvm_warning("CFG", "default used")
'uvm_error("CPU", "mispatch")
'uvm_fatal("TB", "no wif")

Gate rule (INFO / WARNING concept):
message prints if msg.verbosity ≤ configured threshold
example: LOW info shown when threshold is MEDIUM
         HIGH info filtered when threshold is MEDIUM

Severity mins:
ERROR and FATAL typically print regardless of verbosity threshold

ID filter:
messages carry an ID string ("DRV", "MON", "SD", ...)
set_report_id_action("DRV", UVM_NO_ACTION) // drop family
or use report_catcher to demote / rewrite

Runtime knobs:
+UVM_VERBOSITY=UVM_MEDIUM
comp.set_report_verbosity_level(UVM_HIGH, _phase(run_phase))

Hierarchy in log:
report includes component context (get_full_name)

Common mistakes:
severity confused with verbosity
uvm_debug spam in every cycle
one ID reused for unrelated messages
expecting errors to be hidden by low verbosity

Legacy reference:
learn_uvm2017_sv_verilator/modules/examples/reporting/reporting.sv

$ grep -n "uvm_info\|uvm_error\|UVM_HIGH\|verbosity"
12: // Learn_uvm2017_sv_verilator/modules/examples/reporting/reporting.sv | head -14
13: * Demonstrates UVM reporting system with severity and verbosity levels.
14: * 3. Master verbosity levels (LOW, MEDIUM, HIGH, FULL, DEBUG)
15: * - 'uvm_info: Informational message (severity: INFO)
16: * - 'uvm_warning: Warning message (severity: WARNING)
17: * - 'uvm_error: Error message (severity: ERROR, continues simulation)
18: * - UVM_HIGH: High detail (debugging)
19: * - Verbosity levels: UVM_LOW, UVM_MEDIUM, UVM_HIGH, UVM_FULL, UVM_DEBUG
20: * - Shows verbosity levels (LOW, MEDIUM, HIGH, FULL, DEBUG)
21: // 'uvm_info(id, message, verbosity)
22: // - verbosity: Verbosity level (UVM_LOW, UVM_MEDIUM, UVM_HIGH, etc.)
23: 'uvm_info("REPORTING", "This is an INFO message (severity: INFO)", UVM_MEDIUM)
24: // HIGH verbosity INFO message
25: // - UVM_HIGH: Only shown when verbosity ≥ UVM_HIGH
26: // - Can be filtered: +UVM_VERBOSITY=UVM_HIGH
```

Real shell — reporting sketch

Pitfalls to watch

- Do not confuse severity with verbosity, severity is how serious
- Do not expect uvm error to respect a low verbosity threshold, it should still print
- Do not flood sim with uvm debug on every cycle, raise threshold for regression runs
- Do not reuse one ID for unrelated messages, you lose filter precision
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On real UVM, sketch one uvm info line with ID and verbosity plus one uvm error
- When you are ready, take the short quiz, then continue to the protocol checker module next